

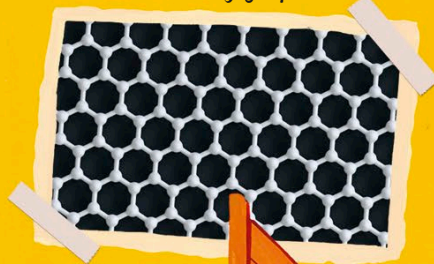
Materials scientists are explorers and inventors. They create advanced materials that help to make faster and more eco-friendly vehicles, longer-lasting batteries, more advanced computers, and better communication devices, such as mobile phones. Here are some world-changing materials and the problems they can solve—or cause.

MATERIALS

Nanomaterials

Something extremely small, such as an atom, can be described as “nano.” Graphene is just one atom thick, but incredibly strong. In addition to being the lightest known nano material, it is also the best conductor of electricity.

Structure of graphene



This dress is made of graphene—however, nanomaterials are usually used for very small parts inside objects.

New from old

“Rare-earth” metals were discovered hundreds of years ago, but are still being used to create new materials. They made many modern devices possible—for example, mobile phones vibrate using rare-earth neodymium magnets!

Tough ceramics

Everyday ceramics come in the form of plates and mugs. Yet, ceramics are also some of the toughest materials in the world. Silicon nitrides can keep their form in temperatures that would melt diamonds! They are used to make parts of machines that need to be very strong.

